

Low-Resource Filipino Adaptation of OmniVoice for Voice Cloning and Text-to-Speech

Research Project Proposal

Current Trends

May 20, 2026

Abstract

Voice cloning is an emerging artificial intelligence technology that can synthesize speech in a target speaker’s voice from short reference audio. Although recent multilingual text-to-speech systems support many languages, practical voice cloning workflows remain strongest in high-resource languages. This proposal investigates how much Filipino-specific fine-tuning improves the pretrained OmniVoice model using curated Filipino read speech from the UP-DSP Philippine Languages Database. The proposed study compares the base OmniVoice model with a Filipino fine-tuned checkpoint and measures the size of the change across intelligibility, naturalness, and speaker similarity metrics. The expected output is a reproducible Filipino fine-tuning workflow, generated audio samples, and an evaluation report suitable for a course-scale research project.

1 Introduction

Voice cloning and text-to-speech (TTS) systems have rapidly improved due to large-scale neural speech models. These systems can generate synthetic speech from text and, in zero-shot voice cloning settings, imitate a speaker’s voice from a short reference recording. This capability has practical applications in education, accessibility, localization, narration, digital assistants, and language learning.

However, most high-quality voice cloning systems are built and evaluated primarily on high-resource languages such as English and Chinese. Filipino and other Philippine languages remain relatively low-resource in speech AI, which limits the quality and availability of practical Filipino voice generation tools. Instead of training a Filipino voice cloning model from scratch, this project proposes adapting an existing multilingual model: OmniVoice.

OmniVoice is a massively multilingual zero-shot TTS model that supports more than 600 languages. It includes Filipino, but the model’s language table lists only about 7.71 hours of Filipino in its training mixture. This creates a research opportunity: measure how much Filipino-specific fine-tuning using a larger curated Filipino speech corpus changes the model’s Filipino generation quality.

2 Research Problem

The research problem is that Filipino voice cloning remains underrepresented in practical open-source workflows, despite the availability of broad multilingual TTS models. OmniVoice supports Filipino, but language support alone does not guarantee optimal performance for Filipino pronunciation, rhythm, naturalness, or speaker similarity.

The proposed study asks how much a low-resource adaptation approach can improve Filipino speech generation without training a model from scratch. This is appropriate for a course-scale

research project because it is experimentally measurable, technically current, and feasible using the existing OmniVoice fine-tuning pipeline.

3 Research Question and Objectives

3.1 Main Research Question

How much does Filipino-specific fine-tuning improve OmniVoice performance for Filipino TTS and voice cloning across intelligibility, naturalness, and speaker similarity metrics?

3.2 Sub-Questions

1. How much does fine-tuning reduce Filipino character error rate or word error rate?
2. How much does fine-tuning change objective and perceived naturalness of generated Filipino speech?
3. How much does fine-tuning change speaker similarity in voice cloning?

3.3 Objectives

1. Prepare a clean Filipino speech-text dataset suitable for OmniVoice fine-tuning.
2. Fine-tune the pretrained `k2-fsa/OmniVoice` checkpoint on Filipino read speech.
3. Generate Filipino speech samples using both the base and fine-tuned models.
4. Evaluate intelligibility, naturalness, and speaker similarity using objective and human evaluation methods.
5. Produce a reproducible workflow and demo samples that can be connected to the existing voice cloning application.

4 Related Work

Recent speech generation research has shifted toward large multilingual and zero-shot systems. Neural codec language models such as VALL-E showed that discrete speech tokens can be used for zero-shot TTS. Voicebox demonstrated large-scale text-guided multilingual speech generation. F5-TTS and E2 TTS explored efficient non-autoregressive TTS using flow matching, while MaskGCT used masked generative codec modeling for zero-shot TTS.

OmniVoice builds on this line of work by using a diffusion language model-style architecture for multilingual zero-shot speech generation. It directly maps text, optional language and instruction tags, and optional reference audio prompts into acoustic tokens. Unlike training a separate speaker model, OmniVoice supports zero-shot voice cloning from a short reference recording.

For Filipino data, the UP-DSP Philippine Languages Database (PLD) provides a relevant corpus for low-resource Philippine speech technology. The PLD paper reports 454:49:43 total duration across Philippine languages, including 48:56:36 of Filipino speech. This makes it suitable for an academic adaptation experiment.

5 Selected Model: OmniVoice

OmniVoice was selected because it is open-source, multilingual, supports voice cloning, and provides official fine-tuning scripts. The model supports more than 600 languages and includes Filipino with language ID `fil`. Figure 1 shows the OmniVoice paper and Figure 2 shows the Filipino entry in the supported-language table.

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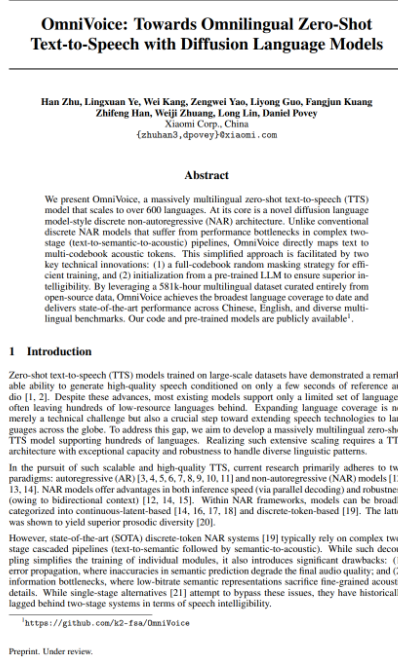


Figure 1: OmniVoice paper first page.

5.1 Architecture

For this project, the key OmniVoice mechanism is its masked acoustic-token generation process:

1. Text is encoded using a language-model tokenizer.
2. Reference audio is converted into discrete acoustic tokens.
3. The model receives text, language tags, and optional reference voice tokens.
4. During training, some acoustic tokens are masked.
5. The model learns to reconstruct the masked acoustic tokens.
6. During inference, generated acoustic tokens are decoded into speech audio.

Fine-tuning is expected to adapt the model’s text-to-acoustic-token behavior toward Filipino speech patterns.

D Supported Languages of OmniVoice

OmniVoice supports **646 languages** with a total of **581k hours** of training data. Detailed language, along with its OmniVoice language ID, ISO 639-3 code, and training data duration are shown in Table 11.

Table 11: Supported languages of OmniVoice along with its OmniVoice language ID, ISO 639-3 code, and training data duration (hours).

Language	ID	ISO	Hours	Language	ID	ISO	Hours	Language	ID	ISO	Hours
Abkhaz	abx	abk	9.71	Abkhazian	abx	abk	9.71	Abkhaz	abx	abk	9.71
Abua	abu	abu	10.27	Adamawa Fulfulde	fub	fub	13.12	Adyghes	ady	ady	32.6
Alade	aal	aal	10.19	Afrikaans	af	afri	4.4	Agwagwune	agw	agw	8.26
Aja (Berin)	ajg	ajg	5.63	Akeba	ken	ken	9.1	Alago	ala	ala	11.04
Albanian	sq	sqi	8.59	Algerian Arabic	arq	arq	9.64	Algerian Saharan Arabic	aao	aao	2.02
Ambo-Puoco Quechua	qva	qva	9.59	Ambo-Puoco Malay	als	als	10.03	Ando Tibetan	ads	ads	56.94
Amharic	am	amh	12.83	Anauag	anw	anw	9.65	Angika	ang	ang	10.65
Antankaranga Malagasy	xmv	xmv	17.9	Angemese	ang	ang	16.4	Arbereshe Albanian	ase	ase	6.11
Arcapi-La Unión Quechua	qua	qua	10.12	Armenian	hy	hye	42.15	Ashle	als	als	10.62
Ashéninka Perené	prq	prq	7.16	Askyan	civ	civ	10.44	Assamese	as	asm	270.85
Astorian	ast	ast	8.48	Astyan	tay	tay	7.02	Awaak	awo	awo	10.22
Ayacucho Quechua	qay	qay	0.05	Azerbaijani	az	aze	9.84	Baatonum	bba	bba	10.53
Bacama	bey	bey	9.94	Bale	bde	bde	9.89	Bala	bal	bal	16.43
Bafut	bfd	bfd	9.03	Bagirmi Fulfulde	fui	fui	15.04	Bago-Kasuntu	bkg	bkg	8.86
Bahama Arabic	abv	abv	10.41	Bakoko	bhb	bhb	6.0	Bakanta-Ganja	bjt	bjt	9.41
Bali	bfi	bfi	16.28	Baksey	bse	bse	9.9	Bamun	bax	bax	10.24
Bangwile	bji	bji	10.0	Banjar	bjp	bjp	11.68	Baniwa	abn	abn	11.2
Basile	bci	bci	10.21	Bara Malagasy	bbr	bbr	12.14	Barok	bjk	bjk	10.16
Basa (Cameroon)	bas	bas	10.66	Basa (Nigeria)	baw	baw	10.27	Basikir	ba	bak	249.1
Basque	eu	eus	479.86	Bata-Mandailing	bem	bem	11.09	Batanga	bun	bun	15.01
Batari	bey	bey	9.8	Bats	bbl	bbl	11.22	Bayet	bda	bda	9.47
Bebele	bsh	bsh	7.52	Belarussian	be	bel	109.43	Bezugli	bex	bex	271.26
Betawi	bew	bew	11.15	Bhili	bhb	bhb	9.98	Bhojpuri	bho	bho	10.05
Bhar	bvf	bvf	10.44	Bima	bip	bip	10.67	Boko	bex	bex	231.57
Boghom	bux	bux	10.48	Bokyi	bky	bky	9.85	Boma	bmj	bmj	10.68
Bondeli	bou	bou	9.98	Borgo Fulfulde	bue	bue	20.1	Bosnian	bos	bos	690.73
Brabai	bth	bth	19.89	Botj	bta	bta	10.68	Breton	bre	bre	25.48
Bradama	bdm	bdm	10.17	Buginese	bug	bug	11.09	Bukharic	bkh	bkh	11.38
Bulgarian	bg	bul	2190.76	Bura-Pahir	bwr	bwr	10.4	Bundeli	bns	bns	10.88
Bunum	bun	bun	9.26	Burak	buk	buk	9.14	Burak	bys	bys	9.92
Burmes	my	mya	12.14	Burmeskuiki	cky	cky	8.96	Cacalotzque Mixtec	mta	mta	9.18
Cajamambo North Lima Quechua	qvi	qvi	9.95	Cakfem-Mashere	yuc	yuc	1302.38	Cameroon Pidgin	wes	wes	10.06
Campanese Sardinian	ceb	ceb	10.16	Canimese	cen	cen	9.85	Catalan	cat	cat	2358.6
Cebuano	ceb	ceb	12.17	Can	can	can	9.85	Central Kurdish	ckb	ckb	137.52
Central Nahuatl	nln	nln	9.51	Central Pame	pbe	pbe	9.69	Central Pado	gst	gst	11.4
Central Pado Nahuatl	nca	nca	9.86	Central Tarahumara	shu	shu	2.29	Central Yopi	csu	csu	2.18
Central-Eastern Niger Fulfulde	fnq	fnq	9.28	Chadian Arabic	shu	shu	2.29	Chichewa	ny	nya	10.8
Chickasaw Zapotec	rpo	rpo	9.85	Chiga	cgg	cgg	10.84	Chimelap Zoque	zoh	zoh	9.35
Chimborazo Highland Quichua	qng	qng	10.12	Chinese	zh	cmn	111343.3	Chiquian Ancestral Quechua	qya	qya	9.99
Chinwasa Tharu	the	the	10.06	Chokwe	ckk	ckk	11.01	Chuvash	cv	cv	23.96
Chuk	ckl	ckl	10.91	Coastal Kono	kjc	kjc	10.18	Copinali Zoque	zoc	zoc	10.07
Cornish	kw	cor	12.15	Cowego Ancestral Quechua	qwa	qwa	9.72	Croatian	hr	hr	2795.31
Cross River Mbembe	mfn	mfn	10.03	Cuyamelco Mixtec	xtu	xtu	9.4	Czech	ces	ces	148.13
Dadaya	dbd	dbd	9.61	Dagbani	dag	dag	10.14	Daneti	dnd	dnd	9.18
Danish	da	dan	1665.98	Dagwa	dag	dag	1.72	Dargwa	dzg	dzg	9.86
Deccan	dec	dec	10.38	Dagwa	dag	dag	11.07	Dera (Nigeria)	kna	kna	11.91
Digwede	dgh	dgh	9.85	Dakki	dkk	dkk	8.83	Dhotchi	dh	dh	38.61
Dhefari Arabic	adf	adf	0.31	Digim-Bwili	dgy	dgy	10.32	Dogri	dgo	dgo	117.04
Domaiki	dok	dok	6.38	Diyali	dyy	dyy	10.85	Duala	dul	dul	12.13
Dutch	nl	nl	2264.13	Diyala	ldl	ldl	11.31	Dyula	dya	dya	0.34
Eastern Baluchi	bgo	bgo	10.58	Diyala	ldl	ldl	11.31	Eastern Egyptian Bedawi Arabic	yer	yer	1.86
Eastern Krahn	kro	kro	9.28	Eastern Mari	mhr	mhr	272.31	Eastern Yiddish	yid	yid	18.43
Ehele	ehb	ehb	1.5	Eggon	ego	ego	9.95	Egyptian Arabic	arz	arz	23.23
Egham	etu	etu	10.3	Egime	elo	elo	11.27	Eloyi	alo	alo	11.23
Enbu	ebu	ebu	9.81	English	en	eng	206061.1	Eryza	myv	myv	3.1
Eton	ish	ish	10.05	Esperanto	eo	epo	1396.64	Esan	esi	esi	960.37
Eton (Cameroon)	eto	eto	7.43	Ewondo	ewo	ewo	12.71	Extremaduran	ext	ext	13.59
Fang (Equatorial Guinea)	fan	fan	3.51	Fai	fai	fai	11.38	Faerle	gar	gar	9.24
Fa'le	fmp	fmp	9.86	Fil	fil	fil	7.71	Filomena Mata-Coubatán Totomac	flp	flp	11.35
Fanti	fti	fti	868.62	Filipino	fil	fil	7.71	French	fra	fra	25875.32
Fulah	ftl	ftl	13.84	Filipino	fil	fil	7.71	Gambian Wolof	wof	wof	9.46
Ganda	lg	lug	447.82	Galician	gl	glg	208.81	Gauar-Bati	gwt	gwt	12.16
Gawri	gwc	gwc	10.83	Gharwali	ghm	ghm	19.14	Ghar	ghy	ghy	12.59
Geji	gyz	gyz	10.49	Ghasi	gbi	gbi	12.12	Georgian	ka	kat	156.96
German	de	deu	21927.13	Gen	goi	goi	5.39	Gheg Albanian	arb	arb	1.92
Ghomali	gbi	gbi	7.32	Gen	goi	goi	5.39	Gilwa	glw	glw	10.51
Gosa Kankani	gon	gon	9.82	Gidar	gid	gid	10.06	Giswani	gsw	gsw	10.0
Gola	gol	gol	9.26	Gidari	gid	gid	10.06	Giswani	gsw	gsw	10.0
Gualdi-Gua	guf	guf	12.21	Gidari	gid	gid	10.06	Giswani	gsw	gsw	10.0
Gujari	guz	guz	8.66	Gidari	gid	gid	10.06	Giswani	gsw	gsw	10.0
Gusi	guz	guz	9.5	Gidari	gid	gid	10.06	Giswani	gsw	gsw	10.0
Gulid Zapotec	guz	guz	9.5	Gidari	gid	gid	10.06	Giswani	gsw	gsw	10.0
				Gidari	gid	gid	10.06	Giswani	gsw	gsw	10.0

Figure 2: OmniVoice supported-language table excerpt showing Filipino.

6 Dataset

6.1 Data Source

The primary dataset is the UP-DSP Philippine Languages Database (UP-DSP-PLD), released through Mozilla Data Collective. The dataset is distributed as WAV audio and LOG transcript files under a CC-BY-NC-4.0 license. This makes it appropriate for academic research but not for commercial deployment.

6.2 Filipino Subset

The Filipino subset is large enough for a small fine-tuning study while remaining feasible for Kaggle-based experimentation.

Table 1: Filipino subset statistics from the PLD paper.

Detail	Value
Language ID	f11
Speakers	135
Utterances	52,879
Total duration	48:56:36
Average utterance length	3.58 seconds



Figure 3: UP-DSP PLD release page on Mozilla Data Collective.

Language	Gender	Speaker Count	Utterance Count	Audio Duration		Tokens	
				Total (h:m:s)	Average (s)	Total	Unique
Bikolano (bik)	F	121	39,260	60:55:58	5.5873	321,721	16,049
	M	85	27,684	40:17:36	5.2397	206,642	14,631
	all	206	66,944	101:13:35	5.4436	528,363	17,005
Cebuano (ceb)	F	86	34,956	35:47:01	3.6852	144,882	8,026
	M	66	27,477	27:44:58	3.6357	114,563	7,267
	all	152	62,433	63:31:59	3.6634	259,445	6,844
English (eng)	F	23	3,156	4:31:30	5.1617	29,376	4,363
	M	7	888	1:01:40	4.1675	8,050	1,483
	all	30	4,044	5:33:11	4.9434	37,426	4,729
Filipino (fil)	F	79	30,617	31:43:55	3.7311	205,346	10,861
	M	56	22,262	20:50:48	3.3712	138,088	7,994
	all	135	52,879	48:56:36	3.5796	343,434	11,481
Hiligaynon (hil)	F	48	17,079	21:49:43	4.6012	99,087	5,397
	M	43	14,908	19:21:58	4.6766	84,676	4,906
	all	91	31,987	41:11:42	4.6363	183,763	5,767
Ilokano (ilo)	F	64	15,429	25:46:37	6.0145	131,316	11,603
	M	60	14,513	25:44:43	6.3862	130,500	11,642
	all	124	29,942	51:31:20	6.1947	261,816	13,270
Kapampangan (pam)	F	104	35,024	49:42:02	5.1086	225,595	12,827
	M	83	26,926	40:37:22	5.4313	176,947	12,629
	all	187	61,950	90:19:25	5.2488	402,542	14,221
Pangasinan (pag)	F	12	3,959	6:01:36	5.4802	24,819	4,302
	M	6	1,945	3:07:00	5.7687	11,698	3,148
	all	18	5,904	9:08:36	5.5753	36,517	4,773
Tausug (tsg)	F	4	1,185	1:43:09	5.2236	12,684	2,023
	M	9	2,103	3:06:34	5.3233	7,279	1,536
	all	13	3,288	4:49:45	5.2874	19,963	2,376
Waray-Waray (war)	F	48	15,337	22:51:12	5.3643	94,764	6,071
	M	26	8,500	12:04:10	5.1118	52,704	5,518
	all	74	23,837	34:55:23	5.2743	147,468	6,291
Total	-	1,030	343,208	454:49:43	4.7708	2,220,737	-

Table 2: Summary statistics for the Philippine Languages Database. Below the each Philippine language name is its language ID in parenthesis, based from the ISO 639-3 standard, as published in Ethnologue (Eberhard et al., 2024) Note that the total token and unique token counts do not include yet the transcripts from the spontaneous speech part as this part of the corpora is still being transcribed.

Figure 4: PLD paper table showing language-level corpus statistics.

6.3 Read Speech Selection

The first experiment will use Filipino read speech. Read speech is preferred because the speaker reads a prepared prompt, so the transcript is already matched to the spoken sentence. Spontaneous speech, where a speaker answers freely, may require stricter verification because the prompt or question may not equal the actual spoken transcript.

7 Methodology

7.1 Data Preparation

The data preparation process will convert Filipino WAV and LOG files into the JSONL format expected by OmniVoice:

```
{  
  "id": "fil_000001",  
  "audio_path": "/path/audio.wav",  
  "text": "Magandang umaga sa inyong lahat.",  
  "language_id": "fil"  
}
```

The preparation steps are:

1. Extract the Filipino subset from the PLD archive.
2. Parse LOG files to pair each WAV file with its transcript.
3. Keep read speech first.
4. Filter samples to 1–10 seconds.
5. Remove empty transcripts, invalid audio files, and placeholder transcripts.
6. Apply light text normalization, such as trimming whitespace and removing control characters.
7. Split the data into train, development, and test sets.

7.2 Fine-Tuning Pipeline

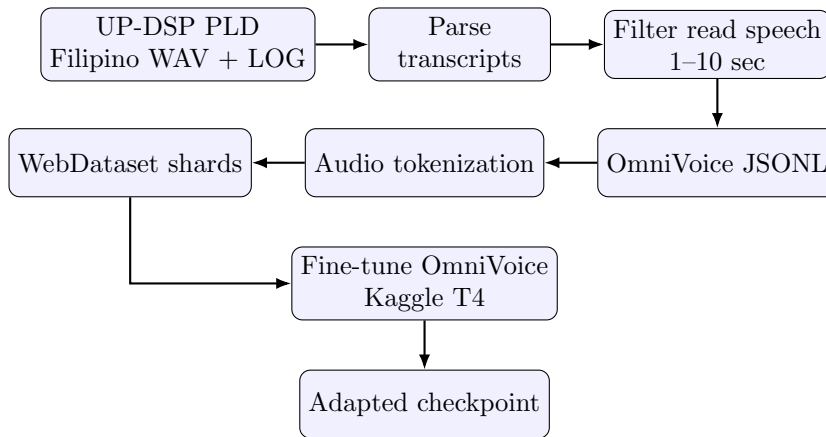


Figure 5: Proposed fine-tuning pipeline.

Audio will be tokenized using the OmniVoice audio tokenizer into WebDataset shards. Fine-tuning will start from the pretrained `k2-fsa/OmniVoice` checkpoint.

7.3 Training Environment

The target training environment is Kaggle with 2x NVIDIA T4 GPUs. Because T4 GPUs have limited memory and do not benefit from BF16 in the same way as newer GPUs, the initial configuration will use FP16, SDPA attention, smaller token batches, and DeepSpeed ZeRO-2.

Table 2: Initial fine-tuning configuration.

Setting	Planned Value
Starting checkpoint	k2-fsa/OmniVoice
Training data size	5–10 hours clean Filipino read speech
Precision	FP16
Attention implementation	SDPA
Learning rate	5e-6
Training steps	1,000–2,000
Optimization	DeepSpeed ZeRO-2

8 Evaluation Plan

The study will compare two systems: the base OmniVoice checkpoint and the Filipino fine-tuned OmniVoice checkpoint. Both systems will generate speech using the same Filipino test prompts and reference voices. The goal is to quantify the size of the change for each metric, not only to state that fine-tuning helped.

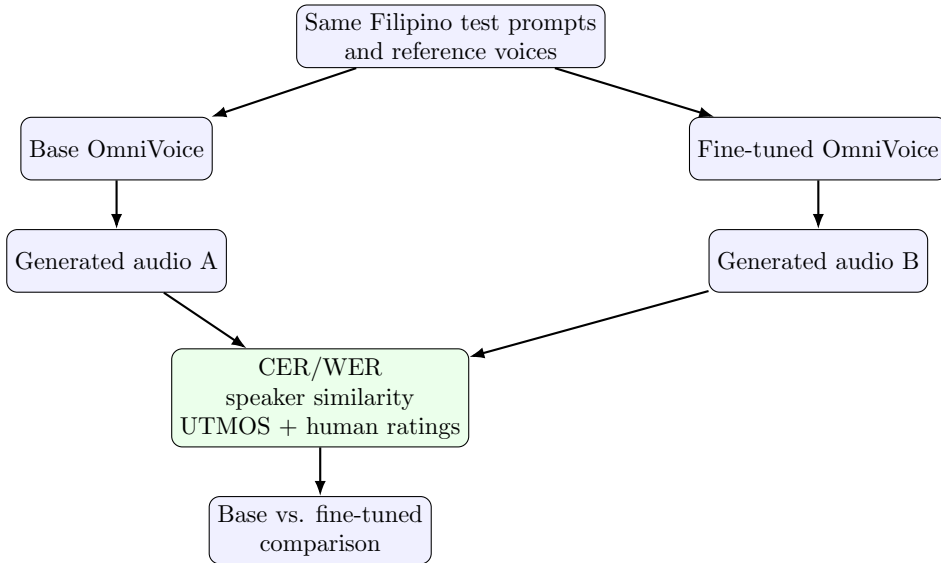


Figure 6: Evaluation design.

8.1 Metrics

- **CER/WER for intelligibility:** generated audio will be transcribed by an automatic speech recognition (ASR) model and compared against the original target text. Character error rate (CER) counts character-level insertions, deletions, and substitutions; word error rate (WER) does the same at the word level. Lower values mean that the generated speech is easier for an ASR system to recognize as the intended sentence. For Filipino, CER may be useful because it is less sensitive to tokenization and word-boundary differences.

- **Speaker similarity for voice cloning:** reference audio and generated audio will be converted into speaker embeddings, then compared using cosine similarity. This follows the same idea as OmniVoice’s SIM-o evaluation, which uses a WavLM-based ECAPA-TDNN speaker model. Higher similarity means the generated speech is closer to the reference speaker’s voice.
- **UTMOS for objective naturalness:** if feasible, UTMOS will be used as an automatic prediction of speech naturalness. It estimates how natural or human-like a generated waveform sounds. Higher scores indicate more natural audio.
- **Human evaluation for listener judgment:** Filipino listeners will rate pronunciation correctness, naturalness, intelligibility, speaker similarity, and overall preference. This is important because ASR-based metrics may miss Filipino pronunciation issues, prosody, accent, or unnatural rhythm.

These metrics are aligned with the OmniVoice paper. OmniVoice evaluates intelligibility using WER or CER depending on the benchmark language, speaker similarity using SIM-o with WavLM-based ECAPA-TDNN embeddings, and objective naturalness using UTMOS. The paper also supplements objective results with human ratings such as CMOS for relative quality and SMOS for speaker similarity to prompt audio. This project adapts the same evaluation logic to the Filipino fine-tuning setting.

9 Scope and Delimitations

The study is limited to Filipino and does not attempt to support all Philippine languages. It fine-tunes an existing model rather than training from scratch. The first dataset subset will be read speech, not spontaneous speech. The work is a research prototype and will not claim commercial readiness or state-of-the-art performance.

10 Ethical and Legal Considerations

Voice cloning can be misused for impersonation or fraud, so generated outputs will be used only for academic demonstration and evaluation. The PLD dataset is licensed under CC-BY-NC-4.0 and is therefore restricted to non-commercial use. The project will not attempt to identify speakers or redistribute the dataset.

11 Expected Contribution

The expected contribution is a practical workflow for adapting OmniVoice to Filipino, including cleaned JSONL manifests, a reproducible Kaggle fine-tuning setup, generated audio samples, and a base-vs-fine-tuned evaluation. The study will provide evidence on whether low-resource Filipino adaptation is feasible under limited compute.

12 Timeline

13 References

- Chen, S., Wang, C., Wu, Y., Zhang, Z., Zhou, L., Liu, S., Chen, Z., Liu, Y., Wang, H., Li, J., et al. (2025). *Neural codec language models are zero-shot text to speech synthesizers*. IEEE/ACM Transactions on Audio, Speech, and Language Processing.

Table 3: Proposed project timeline.

Period	Task
Day 1–2	Download and inspect PLD; parse Filipino WAV and LOG files.
Day 3	Clean, filter, and split Filipino read speech; create JSONL manifests.
Day 4	Set up Kaggle environment and run tokenization smoke test.
Day 5–6	Fine-tune OmniVoice on 5–10 hours of Filipino read speech.
Day 7	Generate base and fine-tuned audio samples.
Day 8	Compute objective metrics and prepare human listening form.
Day 9–10	Summarize results, update demo samples, and write final report.

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